

Seminar 9

NP-completeness

1. Give a non-deterministic polynomial-time algorithm for the following computational problems:
 - (a) SUBSET-SUM;
 - (b) VERTEX-COVER;
 - (c) 3-COLORABILITY;
 - (d) HAMILTONIAN-PATH;
 - (e) HAMILTONIAN-CYCLE;
 - (f) GRAPH ISOMORPHISM.
2. Prove that the 2-colorability problem is in P.
3. Prove that the 2-SAT problem is in P.